Q1.

Matthew measured the pH of different soils.

(a) Tick **one** box in each row to show if each soil is acidic, neutral **or** alkaline.

soil	pH of soil	acidic	neutral	alkaline
A	4.5			
B	5.5			
C	6.3			
D	7.0			
E	7.8			

2 marks

(b) A hydrangea is a flowering plant. Matthew notices that the colour of hydrangea flowers is different for plants grown in different places.

He records the colour of the flowers on each plant.

His results are shown in the table below.



hydrangea flower

soil	pH of soil	colour of flowers			
		blue	violet	light pink	dark pink
A	4.5	✓			
B	5.5		✓		
C	6.3		✓		
D	7.0			✓	
E	7.8				✓

Look at Matthew's results.
Do his results support the statement that the colour of hydrangea flowers depends on pH?

yes ☐ no ☐

Explain your answer.

.....

.....

1 mark

(c) Matthew measured the pH of the soil near hydrangea plants found in different places.

Suggest one **other** variable Matthew could **not** control in his investigation.

.....

.....

1 mark

(d) Matthew wants to find out if the colour of blue hydrangea flowers depends on inherited factors **or** environmental factors.
The flowers were growing in soil of pH 4.5.
He plants them in soil of pH 6.3.

Complete the table below to show the colours of the new flowers in soil of **pH 6.3**

(i) if the colour is due to inheritance
(ii) if the colour is due to the environment

Use the table above to complete the table below.

	colour
starting colour of hydrangea flowers	blue
colour of new flowers if only due to inheritance	
colour of new flowers if only due to environment	

2 marks
maximum 6 marks

Q1.

(a)

soil	pH of soil	acidic	neutral	alkaline
A	4.5	✓		
B	5.5	✓		
C	6.3	✓		
D	7.0		✓	
E	7.8			✓

*for all **five** ticks correct, award two marks*
*for any **three** or **four** ticks correct, award one mark*
*for any **one** or **two** ticks correct, award no marks*
if more than one tick is placed in any one row, do not credit that row

2 (L5)

(b) • yes ✓
as pH increases the flowers become darker pink **or** less blue

*accept 'as pH **or** acidity changes the flower colour changes'*
accept 'acid soil has blue flowers and alkaline soil has pink flowers'
***both** the tick box **and** a consistent explanation are needed for the mark*

or

• no ✓

any **one** from

• there is no change between pH 5.5 – 6.3

• there is insufficient data

1 (L6)

(c) any **one** from

• type of hydrangea
accept 'type of plant'
*accept 'size **or** age of plant'*

• amount of water **or** rainfall
'how much it is watered' is insufficient

• amount of light **or** shade
accept 'light'

- soil composition

  accept 'type of soil'

  *'the acidity **or** pH of the soil' is insufficient*

- nutrients **or** minerals **or** ions in the soil

  *accept named nutrients, minerals **or** ions*

- weather **or** temperature

  accept 'climate'
  accept 'pH of rain'

1 (L5)

(d)

- blue
- violet

accept 'purple' for violet

2 (L6)

[6]